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References:

Chandrika Prasad, Advance Engineering Mathematics, Stanford Publishers and Distributors

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

THE SCOPE, COURSE OBJECTIVES AND COURSE OUTCOMES OF DMS

LECTURE NO:-1

The Scope, Course Objectives and Course Outcomes of DMS

Scope of Discrete Mathematical Structures

Discrete mathematics is the branch of mathematics dealing with objects that can assume only distinct, separated values. The term "**Discrete Mathematics**" is therefore used in contrast with "**Continuous Mathematics**," which is the branch of mathematics dealing with objects that can vary smoothly (and which includes, for example, calculus). Whereas discrete objects can often be characterized by integers, continuous objects require real numbers.

The study of how discrete objects combine with one another and the probabilities of various outcomes is known as **combinatorics**. Other fields of mathematics that are considered to be part of discrete mathematics include **graph theory** and **the theory of computation**. Topics in **number theory** such as congruence's and recurrence relations are also considered part of discrete mathematics.

The study of topics in discrete mathematics usually includes the **study of algorithms**, their **implementations**, and **efficiencies**. Discrete mathematics is the mathematical language of computer science, and as such, its importance has increased dramatically in recent decades.

The set of objects studied in discrete mathematics can be **finite** or **infinite**. The term finite mathematics is sometimes applied to parts of the field of discrete mathematics that deals with finite sets, particularly those areas relevant to business.

Research in discrete mathematics increased in the latter half of the twentieth century partly due to the development of digital computers which operate in discrete steps and store data in discrete bits. Concepts and notations from discrete mathematics are useful in studying and describing objects and problems in branches of computer science, such as **computer algorithms**, **programming languages**, **cryptography**, **automated theorem proving**, and **software development**. Conversely, computer implementations are significant in applying ideas from discrete mathematics to real-world problems, such as in **operations research**.

IMPORTANCE OF DISCRETE MATHEMATICS IN COMPUTER SCIENCE

Achieving working knowledge of many principles of computer science requires mastery of certain relevant mathematical concepts and skills. For example, A grasp of **Boolean algebra** including **DeMorgans Law** is useful for understanding Boolean expressions and the basics of **combinational circuits** concepts surrounding the growth of functions and summations are useful for analysis of loop control structures exposure to solving recurrence relations is de rigeur for the analysis of recursive algorithms and an introduction to proof methods facilitates consideration of program correctness and thinking rigorously in general.

Students are introduced to proof techniques before they begin to consider the idea of proving programs correct. They learn about propositional logic and Boolean algebra before they study some very elementary circuits and learn decision control structures and Boolean variables. They are introduced to predicate logic near the time they are beginning programming and learning about variables. They learn about growth of functions big-O notation and summations

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before they analyze loops and nested loops and they have the tools to begin algorithm analysis from the time they first begin to learn about iterative constructs. In conjunction with an introduction to number theory they do laboratory and programming exercises involving an assortment of integer algorithms.

Students learn about recursive definitions recurrence relations, analyzing recursive algorithms and writing recursive algorithms and programs together in the same course. They study matrices and matrix manipulations in conjunction with the array data structure. They learn about permutations and combinations, relations, graphs, and trees at the same time that their programming knowledge and sophistication are improving and they can do increasingly interesting programming exercises involving these concepts.

Course Objectives

1. General Objectives: Throughout the course, students will be expected to demonstrate their understanding of Discrete Mathematics by being able to do each of the following:

1.1 Use mathematically correct terminology and notation.

1.2 Construct correct direct and indirect proofs.

1.3 Use division into cases in a proof.

1.4 Use counterexamples.

1.5 Apply logical reasoning to solve a variety of problems.

2. Propositional Logic

2.1 Write English sentences for logical expressions and vice-versa. Use standard notations of propositional logic.

2.2 Complete and use truth tables for expressions involving the following logical connectives: negation, conjunction, disjunction, conditional, and biconditional.

2.3 Define and use the terms: proposition (statement), converse, inverse, contrapositive, tautology, and contradiction.

2.4 Apply standard logical equivalences. Be able to prove that two logical expressions are or are not logically equivalent.

2.5 Determine if a logical argument is valid or invalid. Apply standard rules of inference including (but not limited to) Modus Ponens, Modus Tollens, Transitivity, and Elimination. Recognize fallacies such as the Converse Error and the Inverse Error.

3. Predicate Logic

3.1 Translate between English sentences and symbols for universally and existentially quantified statements, including statements with multiple quantifiers.

3.2 Write the negation of a quantified statement involving either one or two quantifiers.

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3.3 Determine if a quantified statement involving either one or two quantifiers is true or false.

4. Elementary Number Theory

4.1 Construct correct direct and indirect (contradiction and contraposition) proofs involving concepts from elementary number theory such as even and odd integers, rational and irrational numbers, and divisibility.

4.2 Find a counterexample to show that a proposed statement involving concepts from elementary number theory is false.

4.3 State and use the Quotient-Remainder Theorem (Division Algorithm) and use the mod function. 4.4 State and use the Fundamental Theorem of Arithmetic.

4.5 Apply the Euclidean Algorithm.

5. Mathematical Induction

5.1 State the Principle of Mathematical Induction.

5.2 Construct induction proofs involving summations, inequalities, and divisibility arguments.

6. Set Theory

6.1 Use set notation, including the notations for subsets, unions, intersections, differences, complements, cross (Cartesian) products, and power sets.

6.2 Prove that a proposed statement involving sets is true, or give a counterexample to show that it is false. In particular, be able to prove that a set is empty.

6.3 Understand and use the terms cardinality, finite, countably infinite, and uncountably infinite, and determine which of these characteristics is associated with a given set.

7. Combinatorics

7.1 Solve counting problems involving the multiplication rule, permutations, and combinations (with and without replacement). Use standard notation.

7.2 Apply the Addition Rule and the Principle of Inclusion and Exclusion.

7.3 Use Pascal's formula and Pascal's Triangle.

7.4 Apply the Binomial Theorem.

7.5 Apply the Pigeonhole Principle.

7.6 Use the basic ideas of discrete probability.

8. Functions

8.1 Define and use the terms function, domain, codomain, range, image, inverse image (preimage), and composition.

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8.2 State the definitions of one-to-one functions (injections), onto functions (surjections), and one-to-one correspondences (bijections). Determine which of these characteristics is associated with a given function.

8.3 Prove that a given function is one-to-one, or give a counterexample to show that it is not.

8.4 Prove that a given function is onto, or give a counterexample to show that it is not.

8.5 Describe the connection between bijective functions and inverses. Be able to find the inverse of an invertible function.

8.6 Explain the connection between cardinality of sets and one-to-one correspondences, and be able to prove that two sets have the same cardinality.

8.7 Find and prove a big-O estimate of growth for a given function.

9. Relations

9.1 State the definitions of binary relation, reflexive, symmetric, transitive, equivalence relation, equivalence class, class representative, and partition.

9.2 Show that a binary relation on a set is an equivalence relation, or give a counterexample to show that it is not.

9.3 Given an equivalence relation on a set, find the equivalence classes of the relation and show that they form a partition of the set.

9.4 Show that congruence modulo m is an equivalence relation on the integers, and that congruence classes modulo m form a partition of the integers.

Upon completion of this course students should be able to do the following:

- Discuss definitions and diagram strategies for potential proofs in logical sequential order without mathematical symbols (plain English).
- Construct mathematical arguments using logical connectives and quantifiers.
- Verify the correctness of an argument using symbolic logic and truth tables.
- Construct proofs using direct proof, proof by contradiction, and proof by cases, or mathematical induction.
- Solve problems using counting techniques and combinatorics.
- Perform operations on discrete structures such as sets, functions, relations or sequences.
- Solve problems involving recurrence relations and generating functions.
- Construct functions and apply counting techniques on sets in the context of discrete probability.
- Apply algorithms and use definitions to solve problems to proof statements in elementary number theory.

COURSE LEARNING OUTCOMES

Students will be able to:

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1. Write an argument using logical notation and determine if the argument is or is not valid.

2. Demonstrate the ability to write and evaluate a proof or outline the basic structure of and give examples of each proof technique described.

3. Understand the basic principles of sets and operations in sets.

4. Prove basic set equalities.

5. Apply counting principles to determine probabilities.

6. Demonstrate an understanding of relations and functions and be able to determine their properties.

7. Determine when a function is 1-1 and "onto".

8. Demonstrate different traversal methods for trees and graphs.

9. Model problems in Computer Science using graphs and trees.

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

PROPOSITION

LECTURE NO:-2

Proposition

A Proposition or a statement or logical sentence is a declarative sentence which is either true or false.

Example1: The following statements are all propositions:

- Jawaharlal Nehru is the first prime minister of India.
- It rained Yesterday.
- If x is an integer, then x^2 is a +ve integer.

Example2: The following statements are not propositions:

- Please report at 11 a.m. sharp
- What is your name?
- $x^2=13$

Propositional Variables

The lower case letters starting from P onwards are used to represent propositions

Example: p: India is in Asia
q: 2 + 2 = 4

Compound Statements

Statements or propositional variables can be combined by means of logical connectives (operators) to form a single statement called compound statements.

The five logical connectives are:

Symbol	Connective	Name
~	Not	Negation

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$\wedge$	And	Conjunction
$\vee$	Or	Disjunction
$\longrightarrow$	Implies or if...then	Implication or conditional
$\longleftrightarrow$	If and only if	Equivalence or biconditional

Basic Logical Operations

1. Negation: It means the opposite of the original statement. If p is a statement, then the negation of p is denoted by ~p and read as 'it is not the case that p.' So, if p is true then ~ p is false and vice versa.

Example: If statement p is Paris is in France, then ~ p is 'Paris is not in France'.

p	~ p
T	F
F	T

2. Conjunction: It means Anding of two statements. If p, q are two statements, then "p and q" is a compound statement, denoted by p $\wedge$ q and referred as the conjunction of p and q. The conjunction of p and q is true only when both p and q are true. Otherwise, it is false.

p	q	p $\wedge$ q
T	T	T
T	F	F
F	T	F
F	F	F

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3. Disjunction: It means Oring of two statements. If p, q are two statements, then "p or q" is a compound statement, denoted by $p \vee q$ and referred to as the disjunction of p and q. The disjunction of p and q is true whenever at least one of the two statements is true, and it is false only when both p and q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

4. Implication / if-then (⟶): An implication $p \longrightarrow q$ is the proposition "if p, then q." It is false if p is true and q is false. The rest cases are true.

p	q	$p \longrightarrow q$
T	T	T
T	F	F
F	T	T
F	F	F

5. If and Only If (↔): $p \leftrightarrow q$ is bi-conditional logical connective which is true when p and q are same, i.e., both are false or both are true.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Derived Connectors

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1. NAND: It means negation after ANDing of two statements. Assume p and q be two propositions. Nanding of pand q to be a proposition which is false when both p and q are true, otherwise true. It is denoted by $p \uparrow q$.

p	q	$p \vee q$
T	T	F
T	F	T
F	T	T
F	F	T

2. NOR or Joint Denial: It means negation after ORing of two statements. Assume p and q be two propositions. NORing of p and q to be a proposition which is true when both p and q are false, otherwise false. It is denoted by $p \uparrow q$.

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

3. XOR: Assume p and q be two propositions. XORing of p and q is true if p is true or q is true but not both and vice-versa. It is denoted by $\mathbf{p \oplus q}$.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Example1: Prove that $X \oplus Y \cong (X \wedge \sim Y) \vee (\sim X \wedge Y)$.

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Solution: Construct the truth table for both the propositions.

X	Y	X⊕Y	~ Y	~ X	X ∧~ Y	~ X∧Y	(X ∧~ Y)∨(~ X∧Y)
T	T	F	F	F	F	F	F
T	F	T	T	F	T	F	T
F	T	T	F	T	F	T	T
F	F	F	T	T	F	F	F

As the truth table for both the proposition is the same.

1. X ⊕ Y ≅ (X ∧~ Y)∨(~ X∧Y). Hence Proved.

Example2: Show that (p ⊕q) ∨(p↓q) is equivalent to p ↑ q.

Solution: Construct the truth table for both the propositions.

p	q	p⊕q	(p↓q)	(p⊕q)∨ (p↓q)	p ↑ q
T	T	F	F	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	T	T	T

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

CONDITIONAL AND BI-CONDITIONAL STATEMENTS

LECTURE NO:-3

Conditional and Bi-Conditional Statements

Conditional Statement

Let p and q are two statements then "if p then q" is a compound statement, denoted by p→ q and referred as a conditional statement, or implication. The implication p→ q is false only when p is true, and q is false; otherwise, it is always true. In this implication, p is called the hypothesis (or antecedent) and q is called the conclusion (or consequent).

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

For Example: The followings are conditional statements.

1. If a = b and b = c, then a = c.
2. If I get money, then I will purchase a computer.

Variations in Conditional Statement

Contrapositive: The proposition ~q→~p is called contrapositive of p →q.

Converse: The proposition q→p is called the converse of p →q.

Inverse: The proposition ~p→~q is called the inverse of p →q.

Example1: Show that p →q and its contrapositive ~q→~p are logically equivalent.

Solution: Construct the truth table for both the propositions:

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p	q	~p	~q	p →q	~q→~p
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

As, the values in both cases are same, hence both propositions are equivalent.

Example2: Show that proposition q→p, and ~p→~q is not equivalent to p →q.

Solution: Construct the truth table for all the above propositions:

p	q	~p	~q	p →q	q→p	~p→~q
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

As, the values of p →q in a table is not equal to q→p and ~p→~q as in fig. So both of them are not equal to p →q, but they are themselves logically equivalent.

Bi-Conditional Statement

If p and q are two statements then "p if and only if q" is a compound statement, denoted as p ↔ q and referred as a biconditional statement or an equivalence. The equivalence p ↔ q is true only when both p and q are true or when both p and q are false.

p	q	p ↔ q
T	T	T

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T	F	F
F	T	F
F	F	T

For Example: (i) Two lines are parallel if and only if they have the same slope.
(ii) You will pass the exam if and only if you will work hard.

Example: Prove that $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

Solution: Construct the truth table for both the propositions:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Since, the truth tables are the same, hence they are logically equivalent. Hence Proved.

Principle of Duality

Two formulas A_1 and A_2 are said to be duals of each other if either one can be obtained from the other by replacing $\wedge$ (AND) by $\vee$ (OR) by $\wedge$ (AND). Also if the formula contains T (True) or F (False), then we replace T by F and F by T to obtain the dual.

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Note1: The two connectives $\wedge$ and $\vee$ are called dual of each other.
2. Like AND and OR, $\uparrow$ (NAND) and $\downarrow$ (NOR) are dual of each other.
3. If any formula of the proposition is valid, then it's dual of each other.

Equivalence of Propositions

Two propositions are said to be logically equivalent if they have exactly the same truth values under all circumstances.

The table1 contains the fundamental logical equivalent expressions:

Laws of the algebra of propositions

Idempotent laws	(i) $p \vee p \cong p$	(ii) $p \wedge p \cong p$
Associative laws	(i) $(p \vee q) \vee r \cong p \vee (q \vee r)$	(ii) $(p \wedge q) \wedge r \cong p \wedge (q \wedge r)$
Commutative laws	(i) $p \vee q \cong q \vee p$	(ii) $p \wedge q \cong q \wedge p$
Distributive laws	(i) $p \vee (q \wedge r) \cong (p \vee q) \wedge (p \vee r)$	(ii) $p \wedge (q \vee r) \cong (p \wedge q) \vee (p \wedge r)$
Identity laws	(i)$p \vee F \cong p$ (iv) $p \wedge F \cong F$	(ii) $p \wedge T \cong p$ (iii) $p \vee T \cong T$
Involution laws	(i) $\neg\neg p \cong p$	
Complement laws	(i) $p \vee \neg p \cong T$	(ii) $p \wedge \neg p \cong T$
DeMorgan's laws:	(i) $\neg(p \vee q) \cong \neg p \wedge \neg q$	(ii) $\neg(p \wedge q) \cong \neg p \vee \neg q$

Example: Consider the following propositions

1. $\sim p \vee \sim q$ and $\sim (p \wedge q)$.

Are they equivalent?

Solution: Construct the truth table for both

p	q	~p	~q	$\sim p \vee \sim q$	$p \wedge q$	$\sim(p \wedge q)$
T	T	F	F	F	T	F

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T	F	F	T	T	F	T
F	T	T	F	T	F	T
F	F	T	T	T	F	T

Tautologies and Contradiction

Tautologies

A proposition P is a tautology if it is true under all circumstances. It means it contains the only T in the final column of its truth table.

Example: Prove that the statement $(p \longrightarrow q) \leftrightarrow (\sim q \longrightarrow p)$ is a tautology.

Solution: Make the truth table of the above statement:

p	q	$p \rightarrow q$	$\sim q$	$\sim p$	$\sim q \longrightarrow p$	$(p \rightarrow q) \longleftrightarrow (\sim q \longrightarrow p)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

As the final column contains all T's, so it is a tautology.

Contradiction:

A statement that is always false is known as a contradiction.

Example: Show that the statement $p \wedge \sim p$ is a contradiction.

Solution:

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Since, the last column contains all F's, so it's a contradiction.

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Contingency:

A statement that can be either true or false depending on the truth values of its variables is called a contingency.

p	q	$p \rightarrow q$	$p \wedge q$	$(p \rightarrow q) \longrightarrow (p \wedge q)$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	F
F	F	T	F	F

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

PREDICATE LOGIC
LECTURE NO:-4

Predicate Logic

Predicate Logic deals with predicates, which are propositions, consist of variables.

Predicate Logic - Definition

A predicate is an expression of one or more variables determined on some specific domain. A predicate with variables can be made a proposition by either authorizing a value to the variable or by quantifying the variable.

The following are some examples of predicates.

- Consider E(x, y) denote "x = y"
- Consider X(a, b, c) denote "a + b + c = 0"
- Consider M(x, y) denote "x is married to y."

Quantifier:

The variable of predicates is quantified by quantifiers. There are two types of quantifier in predicate logic - Existential Quantifier and Universal Quantifier.

Existential Quantifier:

If p(x) is a proposition over the universe U. Then it is denoted as $\exists x\, p(x)$ and read as "There exists at least one value in the universe of variable x such that p(x) is true. The quantifier $\exists$ is called the existential quantifier.

There are several ways to write a proposition, with an existential quantifier, i.e.,

$(\exists x \in A)p(x)$ or $\exists x \in A$ such that p (x) or $(\exists x)p(x)$ or p(x) is true for some $x \in A$.

Universal Quantifier:

If p(x) is a proposition over the universe U. Then it is denoted as $\forall x, p(x)$ and read as "For every $x \in U, p(x)$ is true." The quantifier $\forall$ is called the Universal Quantifier.

There are several ways to write a proposition, with a universal quantifier.

$\forall x \in A, p(x)$ or p(x), $\forall x \in A$ Or $\forall x, p(x)$ or p(x) is true for all $x \in A$.

Negation of Quantified Propositions:

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When we negate a quantified proposition, i.e., when a universally quantified proposition is negated, we obtain an existentially quantified proposition,and when an existentially quantified proposition is negated, we obtain a universally quantified proposition.

The two rules for negation of quantified proposition are as follows. These are also called DeMorgan's Law.

Example: Negate each of the following propositions:

1. $\forall x\, p(x) \wedge \exists\, y\, q(y)$

Sol: $\sim.\ \forall x\, p(x) \wedge \exists\, y\, q(y))$
$\cong\ \sim \forall\, x\, p(x) \vee \sim\ \exists y q\,(y) \qquad (\because \sim\ (p \wedge q) = \sim\ p \vee \sim\ q)$
$\cong\ \exists\, x \sim p(x) \vee \forall y \sim\ q(y)$

2. $(\exists x \in U)\ (x+6=25)$

Sol: $\sim(\ \exists\ x \in U)\ (x+6=25)$
$\cong\ \forall\ x \in U \sim (x+6)=25$
$\cong\ (\forall\ x \in U)\ (x+6) \neq 25$

3. $\sim(\ \exists\ x\, p(x) \vee \forall\ y\, q(y)$

Sol: $\sim(\ \exists\ x\, p(x) \vee \forall\ y\, q(y))$
$\cong\ \sim \exists\ x\, p(x) \wedge \sim \forall\ y\, q(y) \qquad (\because \sim(p \vee q) = \sim\ p \wedge \sim\ q)$
$\cong\ \forall\ x \sim\ p(x) \wedge \exists y \sim q(y))$

Propositions with Multiple Quantifiers:

The proposition having more than one variable can be quantified with multiple quantifiers. The multiple universal quantifiers can be arranged in any order without altering the meaning of the resulting proposition. Also, the multiple existential quantifiers can be arranged in any order without altering the meaning of the proposition.

The proposition which contains both universal and existential quantifiers, the order of those quantifiers can't be exchanged without altering the meaning of the proposition, e.g., the proposition $\exists x\ \forall\ y\, p(x,y)$ means "There exists some x such that p (x, y) is true for every y."

Example: Write the negation for each of the following. Determine whether the resulting statement is true or false. Assume U = R.

1. $\forall\ x\ \exists\ m(x^2<m)$

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Sol: Negation of $\forall$ x $\exists$ m(x^2<m) is $\exists$ x $\forall$ m ($x^2 \geq m$). The meaning of $\exists$ x $\forall$ m ($x^2 \geq m$) is that there exists for some x such that $x^2 \geq m$, for every m. The statement is true as there is some greater x such that $x^2 \geq m$, for every m.

2. $\exists$ m $\forall$ x(x^2<m)

Sol: Negation of $\exists$ m $\forall$ x (x^2<m) is $\forall$ m $\exists$ x ($x^2 \geq m$). The meaning of $\forall$ m $\exists$ x ($x^2 \geq m$) is that for every m, there exists for some x such that $x^2 \geq m$. The statement is true as for every m, there exists for some greater x such that $x^2 \geq m$.

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

NORMAL FORMS

LECTURE NO:-5

Normal Forms

The problem of finding whether a given statement is tautology or contradiction or satisfiable in a finite number of steps is called the Decision Problem. For Decision Problem, construction of truth table may not be practical always. We consider an alternate procedure known as the reduction to normal forms.

There are two such forms:

1. Disjunctive Normal Form (DNF)
2. Conjunctive Normal Form

Disjunctive Normal Form (DNF): If p, q are two statements, then "p or q" is a compound statement, denoted by $p \vee q$ and referred as the disjunction of p and q. The disjunction of p and q is true whenever at least one of the two statements is true, and it is false only when both p and q are false

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example: - if p is "4 is a positive integer" and q is "$\sqrt{5}$ is a rational number", then $p \vee q$ is true as statement p is true, although statement q is false.

Conjunctive Normal Form: If p, q are two statements, then "p and q" is a compound statement, denoted by $p \wedge q$ and referred as the conjunction of p and q. The conjunction of p and q is true only when both p and q are true, otherwise, it is false

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p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

Example: if statement p is "6<7" and statement q is "-3>-4" then the conjunction of p and q is true as both p and q are true statements.

3) Principle Disjunctive normal form

Suppose, **P** and **Q** are two variables. We construct all possible formulas which consist of a conjunction of **P** or in negation and ~~conjunction of~~ **Q** or its negation. Now of the formulas should contains both a variable and its negation. For two variables **P** and **Q** there is **2^2** such formula.

These formulas are called minterms or Boolean conjunction of **p** and **Q** from the truth tables of theses minterms, it is clear that no two minterms are equivalent. Each minterm has the truth value **T** for exactly one combination of the truth value of the variables **P** and **Q**.

For a given formula an equivalent formula consisting of a disjunction of **minterms** only is known as its principle disjunction normal form. Such a normal form is also said to be the sum-product canonical form.

4) Principle conjunctive normal form

Given a number of variables maxterm of these variables is a formula which consists of disjunction in which each variable or its negations but not both appear only once. Observe that the **maxterm** are the duals of **minterms**. Therefore each of the **maxterm** has the truth value **F** for exactly one combination of the truth values of the variables.

The **principle of conjunctive normal form** or the **product-sum canonical form**, the equivalent formula consists of only the conjunction of the **maxterms** only.

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

LOGICAL IMPLICATION AND LOGICAL EQUAVALENCE

LECTURE NO:-6

If asked to proof equal then by truth table.
and if say by logical equivilance then by formulas

Logical Implication And Logical Equavalence

Suppose there are two compound statements, X and Y, which will be known as logical equivalence if and only if the truth table of both of them contains the same truth values in their columns. With the help of symbol = or ⇔, we can represent the logical equivalence. So X = Y or X ⇔ Y will be the logical equivalence of these statements.

With the help of the logical equivalence definition, we have cleared that if the compound statements X and Y are logical equivalence, in this case, the X ⇔ Y must be Tautology.

Laws of Logical Equivalence

In this law, we will use the 'AND' and 'OR' symbols to explain the law of logical equivalence. Here, AND is indicated with the help of ∧ symbol and OR is indicated with the help of ∨ symbol. There are various laws of logical equivalence, which are described as follows:

Idempotent Law:

In the idempotent law, we only use a single statement. According to this law, if we combine two same statements with the symbol ∧(and) and ∨(or), then the resultant statement will be the statement itself. Suppose there is a compound statement P. The following notation is used to indicate the idempotent law:

1. P ∨ P ? P
2. P ∧ P ? P

The truth table for this law is described as follows:

P	P	P ∨ P	P ∧ P
T	T	T	T
F	F	F	F

This table contains the same truth values in the columns of P, P ∨ P and P ∧ P.

Hence we can say that P ∨ P = P and P ∧ P = P.

Commutative Laws:

The two statements are used to show the commutative law. According to this law, if we combine two statements with the symbol ∧(and) or ∨(or), then the resultant statement will be the same even if we change

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the position of the statements. Suppose there are two statements, P and Q. The proposition of these statements will be false when both statements P and Q are false. In all the other cases, it will be true. The following notation is used to indicate the commutative law:

1. $P \vee Q ? Q \vee P$
2. $P \wedge Q ? Q \wedge P$

The truth table for these notations is described as follows:

P	Q	P ∨ Q	Q ∨ P
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

This table contains the same truth values in the columns of $P \vee Q$ and $Q \vee P$.

Hence we can say that $P \vee Q ? Q \vee P$.

Same as we can prove $P \wedge Q ? Q \wedge P$.

Associative Law:

The three statements are used to show the associative law. According to this law, if we combine three statements with the help of brackets by the symbol ∧(and) or ∨(or), then the resultant statement will be the same even if we change the order of brackets. That means this law is independent of grouping or association. Suppose there are three statements P, Q and R. The proposition of these statements will be false when P, Q and R are false. In all the other cases, it will be true. The following notation is used to indicate the associative law:

1. $P \vee (Q \vee R) ? (P \vee Q) \vee R$
2. $P \wedge (Q \wedge R) ? (P \wedge Q) \wedge R$

The truth table for these notations is described as follows:

P	Q	R	P ∨ Q	Q ∨ R	(P ∨ Q) ∨ R	P ∨ (Q ∨ R)
T	T	T	T	T	T	T

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T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	F	F	F

This table contains the same truth values in the columns of P ∨ (Q ∨ R) and (P ∨ Q) ∨ R.

Hence we can say that P ∨ (Q ∨ R) ? (P ∨ Q) ∨ R.

Same as we can prove P ∧ (Q ∧ R) ? (P ∧ Q) ∧ R

Distributive Law:

The three statements are used to show the distributive law. According to this law, if we combine a statement by the ∨(OR) symbol with the two other statements which are joined with the symbol ∧(AND), then the resultant statement will be the same even if we are separately combining the statements with the symbol ∨(OR) and combining the joined statements with ∧(AND). Suppose there are three statements P, Q and R. The following notation is used to indicate the distributive law:

P ∨ (Q ∧ R) ? (P ∨ Q) ∧ (P ∨ R)

P ∧ (Q ∨ R) ? (P ∧ Q) ∨ (P ∧ R)

The truth table for these notations is described as follows:

P	**Q**	**R**	Q ∧ R	P∨(Q ∧R)	P ∨ Q	P ∨ R	(P ∨ Q) ∧ (P ∨ R)
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T

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F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

This table contains the same truth values in the columns of P ∨ (Q ∧ R) and (P ∨ Q) ∧ (P ∨ R).

Hence we can say that P ∨ (Q ∧ R) = (P ∨ Q) ∧ (P ∨ R)

Same as we can prove P ∧ (Q ∨ R) ? (P ∧ Q) ∨ (P ∧ R)

Identity Law:

A single statement is used to show the identity law. According to this law, if we combine a statement and a True value with the symbol ∨(or), then it will generate the True value. If we combine a statement and a False value with the symbol ∧(and), then it will generate the statement itself. Similarly, we will do this with the opposite symbols. That means if we combine a statement and a True value with the symbol ∧(and), then it will generate the statement itself, and if we combine a statement and a False value with the symbol ∨(or), then it will generate the False value. Suppose there is a compound statement P, a true value T and a false value F. The following notation is used to indicate the identity law:

1. P ∨ T ? T and P ∨ F ? P
2. P ∧ T ? P and P ∧ F ? F

The truth table for these notations is described as follows:

P	T	F	P ∨ T	P ∨ F
T	T	F	T	T
F	T	F	T	F

This table contains the same truth values in the columns of P ∨ T and T. Hence, we can say that P ∨ T = T. Similarly, this table also contains the same truth values in the columns of P ∨ F and P. Hence we can say that P ∨ F = P.

Same as we can prove P ∧ T ? P and P ∧ F ? F

Complement Law:

A Single statement is used in the complement law. According to this law, if we combine a statement with its complement statement with the symbol ∨(or), then it will generate the True value, and if we combine these statements with the symbol ∧(and), then it will generate the False value. If we negate a true value, then it will generate a false value, and if we negate a false value, then it will generate the true value.

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The following notation is used to indicate the complement law:

1. $P \vee \neg P ? T$ and $P \wedge \neg P ? F$
2. $\neg T ? F$ and $\neg F ? T$

The truth table for these notations is described as follows:

P	¬P	T	¬T	F	¬F	P ∨ ¬P	P ∧ ¬P
T	F	T	F	F	T	T	F
F	T	T	F	F	T	T	F

This table contains the same truth values in the columns of $P \vee \neg P$ and T. Hence, we can say that $P \vee \neg P = T$. Similarly, this table also contains the same truth values in the columns of $P \wedge \neg P$ and F. Hence we can say that $P \wedge \neg P = F$.

This table contains the same truth values in the columns of ¬T and F. Hence, we can say that ¬T = F. Similarly, this table contains the same truth values in the columns of ¬F and T. Hence we can say that ¬F = T.

Double Negation Law or Involution Law

A single statement is used to show the double negation law. According to this law, if we do the negation of a negated statement, then the resultant statement will be the statement itself. Suppose there is a statement P and a negate statement ¬P. The following notation is used to indicate the Double negation law:

1. $\neg(\neg P) ? P$

The truth table for these notations is described as follows:

P	¬P	¬(¬P)
T	F	T
F	T	F

This table contains the same truth values in the columns of ¬(¬P) and P. Hence we can say that ¬(¬P) = P.

De Morgan's Law:

The two statements are used to show De Morgan's law. According to this law, if we combine two statements with the symbol ∧(AND) and then do the negation of these combined statements, then the resultant statement will be the same even if we combine the negation of both statements separately with the symbol

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∨(OR). Suppose there are two compound statements, P and Q. The following notation is used to indicate De Morgan's Law:

1. ¬(P ∧ Q) ? ¬P ∨ ¬Q
2. ¬(P ∨ Q) ? ¬P ∧ ¬Q

The truth table for these notations is described as follows:

P	Q	¬P	¬Q	P ∧ Q	¬(P ∧ Q)	¬ P ∨ ¬Q
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

This table contains the same truth values in the columns of ¬(P ∧ Q) and ¬ P ∨ ¬Q. Hence we can say that ¬(P ∧ Q) = ¬ P ∨ ¬Q.

Same as we can prove ¬(P ∨ Q) ? ¬P ∧ ¬Q

Absorption Law:

The two statements are used to show the absorption law. According to this law, if we combine a statement P by ∨(OR) symbol with the same statement P and one other statement Q, which are joined with the symbol ∧(AND), then the resultant statement will be the first statement P. The same result will be generated if we interchange the symbols. Suppose there are two compound statements, P and Q. The following notation is used to indicate the Absorption Law:

1. P ∨ (P ∧ Q) ? P
2. P ∧ (P ∨ Q) ? P

The truth table for these notations is described as follows:

P	Q	P ∧ Q	P ∨ Q	P ∨ (P ∧ Q)	P ∧ (P ∨ Q)
T	T	T	T	T	T
T	F	F	T	T	T

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F	T	F	T	F	F
F	F	F	F	F	F

This table contains the same truth values in the columns of P ∨ (P ∧ Q) and P. Hence we can say that P ∨ (P ∧ Q) ? P.

Similarly, this table also contains the same truth values in the columns of P ∧ (P ∨ Q) and P. Hence we can say that P ∧ (P ∨ Q) ? P.

There are various examples of logical equivalence. Some of them are described as follows:

Example 1: In this example, we will establish the equivalence property for a statement, which is described as follows:

p → q ? ¬p ∨ q

Solution:

We will prove this with the help of a truth table, which is described as follows:

P	Q	¬p	p → q	¬p ∨ q
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

This table contains the same truth values in the columns of p → q and ¬p ∨ q. Hence we can say that p → q ? ¬p ∨ q.

Example 2: In this example, we will establish the equivalence property for a statement, which is described as follows:

P ↔ Q ? (P → Q) ∧ (Q → P)

Solution:

P	Q	P → Q	Q → P	P ↔ Q	(P → Q) ∧ (Q → P)
T	T	T	T	T	T

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T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

This table contains the same truth values in the columns of P ↔ Q and (P → Q) ∧ (Q → P). Hence we can say that P ↔ Q ? (P → Q) ∧ (Q → P).

Example 3: In this example, we will use the equivalent property to prove the following statement:

p ↔ q ? (p ∧ q) ∨ (¬ p ∧ ¬q)

Solution:

To prove this, we will use some of the above-described laws and from this law we have:

p ↔ q ? (¬p ∨ q) ∧ (¬q ∨ p) .....(1)

Now we will use the Commutative law in the above equation and get the following:

? (¬ p ∨ q) ∧ (p ∨ ¬q)

Now we will use the Distributive law in this equation and get the following:

? (¬ p ∧ (p ∨ ¬q)) ∨ (q ∧ (p ∨ ¬q))

Now we will use Distributive law in this equation and get the following:

? (¬ p ∧ p) ∨ (¬p ∧ ¬q) ∨ (q ∧ p) ∨ (q ∧ ¬q)

Now we will use the complement law in this equation and get the following:

? F ∨ (¬p ∧ ¬q) ∨ (q ∧ p) ∨ F

Now we will use the identity law and get the following:

? (¬ p ∧ ¬ q) ∨ (q ∧ p)

Now we will use the Commutative law in this equation and get the following:

? (p ∧ q) ∨ (¬ p ¬q)

Finally, equation (1) becomes the following:

p ↔ q ? (p ∧ q) ∨ (¬ p ¬q)

Finally, we can say that the equation (1) becomes p ↔ q ? (p ∧ q) ∨ (¬ p ∧ ¬q)

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

PROBLEMS ON LOGICAL CONNECTIVES

LECTURE NO:-7

Problems on Logical Connectives :

1. Show that $x \wedge y$ is logically equivalent to $y \wedge x$.

(1)	(2)	(3)	(4)
x	y	$x \wedge y$	$y \wedge x$
F	F	F	F
F	T	F	F
T	F	F	F
T	T	T	T

2. Show that $(a \vee b) \vee c$ is logically equivalent to $a \vee (b \vee c)$.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
a	b	c	$a \vee b$	$(a \vee b) \vee c$	$b \vee c$	$a \vee (b \vee c)$
F	F	F	F	F	F	F
F	F	T	F	T	T	T
F	T	F	T	T	T	T
F	T	T	T	T	T	T
T	F	F	T	T	F	T
T	F	T	T	T	T	T
T	T	F	T	T	T	T
T	T	T	T	T	T	T

3. Given the following statements

(a) $(x \vee y) \wedge \sim y$

(b) $[(a \rightarrow b) \wedge (\sim a \vee b)] \leftrightarrow (b \rightarrow a)$

construct truth tables for each.

(1)	(2)	(3)	(4)	(5)
x	y	$x \vee y$	$-y$	$(x \vee y) \wedge (-y)$

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F	F	F	T	F
F	T	T	F	F
T	F	T	T	T
T	T	T	F	F
		W	Z	W ∧ Z

4. Construct a truth table for the statement a ∧ (b ∨ ~ c)

(1)	(2)	(3)	(4)	(5)	(6)
a	b	c	~ c	b ∨ ~ c	a ∧ (b ∨ c)
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	F
F	F	T	F	F	F
F	F	F	T	T	F

5. Construct a truth table for [(a ∨ b) ∧ ~ b] → a.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
a	b	[(a ∨ b)	∧	~ b]	→	a
T	T	T	F	F	T	T
T	F	T	T	T	T	T
F	T	T	F	F	T	F
F	F	F	F	T	T	F

Conditional And Biconditional Statements

6. Symbolize and determine the validity of the following argument:
"The disjunction of x and y is true, and x is true. Therefore, y

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must be false."

x	y	~ y	x ∨ y	(x ∨ y) ∧ x	[(x ∨ y) ∧ x] → ~ y
T	T	F	T	T	F
T	F	T	T	T	T
F	T	F	T	F	T
F	F	T	F	F	T

7. Verify the following argument, "If he is a college student, then he is intelligent. He is a college student. Therefore, he is intelligent."

p	q	p → q	(p → q) ∧ p	[(p → q) ∧ p] → q
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

8. Is the following a valid argument?
"All dogs are animals.
A horse is an animal.
Therefore, a horse is a dog."

p	q	p → q	(p → q) ∧ q	[(p → q) ∧ q] → p
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

THEOREM PROVING TECHNIQUES

LECTURE NO:-8

Mathematical Induction

Mathematical Induction is a technique of proving a statement, theorem or formula which is thought to be true, for each and every natural number n. By generalizing this in form of a principle which we would use to prove any mathematical statement is '**Principle of Mathematical Induction**'.

For example: 13 +23 + 33 + ….. +n3 = (n(n+1) / 2)2, the statement is considered here as true for all the values of natural numbers.

Principle of Mathematical Induction Solution and Proof

Consider a statement P(n), where n is a natural number. Then to determine the validity of P(n) for every n, use the following principle:

Step 1: Check whether the given statement is true for n = 1.

Step 2: Assume that given statement P(n) is also true for n = k, where k is any positive integer.

Step 3: Prove that the result is true for P(k+1) for any positive integer k.

If the above-mentioned conditions are satisfied, then it can be concluded that P(n) is true for all n natural numbers.

Proof:

The first step of the principle is a *factual statement* and the second step is a *conditional one*. According to this if the given statement is true for some positive integer k only then it can be concluded that the statement P(n) is valid for n = k + 1.

This is also known as the *inductive step* and the assumption that P(n) is true for n=k is known as the *inductive hypothesis.*

Solved problems

Example 1: Prove that the sum of cubes of n natural numbers is equal to ([n(n+1)]/2)2 for all n natural numbers.

Solution:

In the given statement we are asked to prove:

$1^3+2^3+3^3+\cdots +n^3 = ([n(n+1)]/2)^2$

Step 1: Now with the help of the principle of induction in Maths, let us check the validity of the given statement P(n) for n=1.

$P(1)=([1(1+1)]/2)^2 = (2/2)^2 = 1^2 =1$.

This is true.

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Step 2: Now as the given statement is true for n=1, we shall move forward and try proving this for n=k, i.e.,

$1^3+2^3+3^3+\cdots +k^3= ([k(k+1)]/2)^2$.

Step 3: Let us now try to establish that P(k+1) is also true.

$1^3+2^3+3^3+....k^3+(k+1)^3=[k(k+1)/2]^2+(k+1)^3$

$=[(k+1)/2]^2[k^2+4(k+1)]$

$=[(k+1)(k+2)/2]^2$

Example 2: **Show that $1 + 3 + 5 + \dots + (2n-1) = n^2$**

Solution:

Step 1: Result is true for n = 1

That is $1 = (1)^2$ (True)

Step 2: Assume that result is true for n = k

$1 + 3 + 5 + \dots + (2k-1) = k^2$

Step 3: Check for n = k + 1

i.e. $1 + 3 + 5 + \dots + (2(k+1)-1) = (k+1)^2$

We can write the above equation as,

$1 + 3 + 5 + \dots + (2k-1) + (2(k+1)-1) = (k+1)^2$

Using step 2 result, we get

$k^2 + (2(k+1)-1) = (k+1)^2$

$k^2 + 2k + 2 -1 = (k+1)^2$

$k^2 + 2k + 1 = (k+1)^2$

$(k+1)^2 = (k+1)^2$

L.H.S. and R.H.S. are same.

So the result is true for n = k+1

By mathematical induction, the statement is true.

We see that the given statement is also true for n=k+1. Hence we can say that by the principle of mathematical induction this statement is valid for all natural numbers n.

Example 3:

Show that $2^{2n}-1$ is divisible by 3 using the principles of mathematical induction.

To prove: $2^{2n}-1$ is divisible by 3

Assume that the given statement be P(k)

Thus, the statement can be written as P(k) = $2^{2n}-1$ is divisible by 3, for every natural number

Step 1: In step 1, assume n= 1, so that the given statement can be written as

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$P(1) = 2^{2(1)}-1 = 4-1 = 3$. So 3 is divisible by 3. (i.e.3/3 = 1)

Step 2: Now, assume that P(n) is true for all the natural numbers, say k

Hence, the given statement can be written as

$P(k) = 2^{2k}-1$ is divisible by 3.

It means that $2^{2k}-1 = 3a$ (where a belongs to natural number)

Now, we need to prove the statement is true for n= k+1

Hence,

$P(k+1) = 2^{2(k+1)}-1$

$P(k+1)= 2^{2k+2}-1$

$P(k+1)= 2^{2k}. 2^2 – 1$

$P(k+1)= (2^{2k}. 4)-1$

$P(k+1) =3.2k +(22k-1)$

The above expression can be written as

$P(k+1) =3.22k + 3a$

Now, take 3 outside, we get

$P(k+1) = 3(22k + a)= 3b$, where “b” belongs to natural number

It is proved that p(k+1) holds true, whenever the statement P(k) is true.

Thus, 22n-1 is divisible by 3 is proved using the principles of mathematical induction

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UNIT-1 (DISCRETE MATHEMATICAL STRUCTURE)

INTRODUCTION TO PROOFS

LECTURE NO:-9

Methods of Proofs

Mathematical proof is an argument we give logically to validate a mathematical statement. In order to validate a statement, we consider two things: A **statement** and **Logical operators**.

A statement is either true or false but not both. Logical operators are AND, OR, NOT, If then, and If and only if. Coupled with quantifiers like for all and there exists. We apply operators on the statement to check the correctness of it.

Types of mathematical proofs:

- **Proof by cases –**

In this method, we evaluate every case of the statement to conclude its truthiness.

Example: For every integer x, the integer x(x + 1) is even

Proof: If x is even, hence, x = 2k for some number k. now the statement becomes:

- 2k(2k + 1)

which is divisible by 2, hence it is even.

If x is odd, hence x = 2k + 1 for some number k, now the statement becomes:

(2k+1)(2k+1+1) = (2k + 1) 2(k + 1)

which is again divisible by 2 and hence in both cases we proved that x(x+1) is even.

- **Proof by contradiction –**

We assume the negation of the given statement and then proceed to conclude the proof.

Example: Prove that sqrt(2) is irrational

Suppose sqrt(2) is rational.

- sqrt(2) = a/b

for some integers a and b with b != 0.

Let us choose integers a and b with sqrt(2) = a/b, such that b is positive and as small as possible. (Well-Ordering Principle)

a^2 = 2b^2

Since a^2 is even, it follows that a is even.

a = 2k for some integer k, so a^2 = 4k^2

b^2 = 2k^2. Since b^2 is even, it follows that b is even.

Since a and b are both even, a/2 and b/2 are integers with b/2 > 0, and sqrt(2) = (a/2)/(b/2), because (a/2)/(b/2) = a/b.

But it contradicts our assumption b is as small as possible. Therefore sqrt(2) cannot be rational.

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- **Proof by induction –**

The Principle of Mathematical Induction (PMI). Let P(n) be a statement about the positive integer n. If the following are true:

1. P(1),

2. (for all n there exists Z+) P(n) implies P(n + 1),

then (for all n there exists Z+) P(n).

Example: For every positive integer n,

$1 + 2 + \cdots + n = n(n + 1)/ 2$

Proof:

Base case: If n = 1,

$1 + \cdots + n = 1$

And

$n(n + 1)/2 = 11$

- **Inductive step:**

Suppose that for a given n there exists Z+,

$1 + 2 + \cdots + n = n(n + 1)/ 2$ ---- (i) (inductive hypothesis)

Our goal is to show that:

$1 + 2 + \cdots + n + (n + 1) = [n + 1]([n + 1] + 1)/ 2$

i.e. $1 + 2 + \cdots + n + (n + 1) = (n + 1)(n + 2) /2$

Add n + 1 both sides to equation (i), we get,

$1 + 2 + \cdots + n + (n + 1)$

$= n(n + 1)/ 2 + (n + 1)$

$= n(n + 1) /2 + 2(n + 1) /2$

$= (n + 2)(n + 1) /2$

- **Direct Proof –**

when we want to prove a conditional statement p implies q, we assume that p is true, and follow implications to get to show that q is then true.

It is Mostly an application of hypothetical syllogism, $[(p \rightarrow r) \wedge (r \rightarrow q)] \rightarrow (p \rightarrow q)]$

We just have to find the propositions that lead us to q.

Theorem: If m is even and n is odd, then their sum is odd

Proof:

Since m is even, there is an integer j such that m = 2j.

Since n is odd, there is an integer k such that n = 2k+1. Then,

- $m+n = (2j)+(2k+1) = 2(j+k)+1$

Since j+k is an integer, we see that m+n is odd.

Types Of Proofs :

Let's say we want to prove the implication $P \Rightarrow Q$. Here are a few options for you to consider.

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1. Trivial Proof –
If we know Q is true, then P ⇒ Q is true no matter what P's truth value is.
Example –
If there are 1000 employees in a geeksforgeeks organization , then 32 = 9.
Explanation –
Let P : There are 1000 employees in geeksforgeeks organization & Q : 32 = 9.
We know that Q is always true and in the truth table we can see that whenever Q is true, P ⇒ Q is true, whatever the true value of P is.

2. Vacuous Proof –
If P is a conjunction (example : P = A ^ B ^ C)of other hypotheses and we know one or more of these hypotheses is false, then P is false and so P → Q is vacuously true regardless of the truth value of Q.

Example –
If 5 ! = 100, then 3 ! = 6.
Explanation –
Let P : 5 ! = 100, & Q : 3 ! = 6.
We know that P is always false and in the truth table we can see that whenever P is False, P ⇒ Q is true, whatever the truth value of Q is.

3. Direct Proof –
Assume P, then prove Q using inference rules, axioms, definitions, and logical equivalences.
Example –
For all integers p and q, if p and q are odd integers, then p + q is an even integer.
Let P denotes : p and q are odd integers
Q : p + q is an even integer
To Prove : P ⇒ Q
Proof –
As p & q are odd integers, they can be represented as :
Assume : p = 2m + 1 and q = 2n + 1, where m & n are also some integers.
Then : p + q =
= (2m + 1) + (2n +1) (Substitution Law)
= am + 2n + 2 (associative and commutative law for addition)
= 2(m + n + 1) (distributive law)
= Number divisible by 2 & hence an even number.

4. Proof By Contradiction –
We start with the assumption that the hypotheses are correct and the conclusion is incorrect, and we try to find a contradiction.
Proof by contradiction is legitimate because :
¬(P ⋀ ¬Q) is equivalent to P ⇒ Q
If we can prove that (P ⋀ ¬Q) is false, then¬(P ⋀ ¬Q) is true, and the equivalent statement P ⇒ Q is likewise true.

Example –
Let x and y be real numbers. If 5a + 25b = 156, then a or b is not an integer.
Proof –
Let P : 5a + 25b = 156 & Q : a or b is not an integer

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¬Q : a or b is an integer
So , we assume that both a and b are integers (¬Q) ⇒ 5(a + 5b) = 156 (distributive law)
⇒ Since a and b are integers, this implies 156 is divisible by 5.
The integer 156 , however, is anyway not divisible by 5. This contradiction gives the result.
It implies that (P ⋀ ¬Q) is false as P is false
then¬(P ⋀ ¬Q) is true and the equivalent statement P ⇒ Q is likewise true.

5. **Proof by Contrapositive –**
We can prove P ⇒ Q indirectly by showing that ¬Q ⇒ ¬P . Assume ¬Q, and then prove ¬P using inference rules, axioms, definitions, and logical equivalences.

Example : For all integers a and b, if a*b is even, then a is even or b is even.
Proof : We prove the contrapositive of the statement:
Let P : a*b is even & Q : a is an even integer or b is an even integer. Then :
¬P : a*b is odd
¬Q : a and b are odd integers

Say ¬Q is true, i.e., a and b are both odd integers
a = 2m + 1 and b = 2n + 1 ; where m and n are integers.
Then :
a*b= (2m + 1)(2n + 1) (by substitution)
= 4mn + 2m + 2n + 1 (by associative, commutative & distributive laws)
= 2(2mn + m + n) + 1 (by distributive law)

Since a*b is twice an integer (As : 2mn + m + n is also an integer) plus 1, a*b is odd.
So it shows that ¬Q ⇒ ¬P. Hence P ⇒ Q

Q. Prove that : n can be odd if and only if n2 is odd.
Solution.
We must prove two implications in order to prove this statement :

1. If n is odd, n2 is odd
2. If n2 is odd, n is odd

Assume –
P : n is odd & Q : n2 is odd.

1. P ⇒ Q :

We are using direct proof to prove it.
Assume n is an odd integer.
Then : n= 2p+ 1 ; for some integer p.
Then n2= (2p+ 1)2= 4p2+ 4p+ 1 =2(2p2+ 2a) + 1,; which is 2*(some integer) + 1.
Thus, we can say that n2 is odd. Thus P ⇒ Q

2. Q ⇒ P :
We are using contra-positive proof way here.
¬Q : n2 is even and ¬P : n is even.
We need to prove : ¬P ⇒ ¬Q (¬P ⇒ ¬Q means that Q ⇒ P)
Assume n is an even integer,
Then n= 2 ; for some integer p.
Then n2= (2p)2= 4p2= 2(2p2), which is an even integer as it is divisible by 2.

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From (1.) P ⇒ Q & from (2) Q ⇒ P, n can be odd if and only if n2 is odd.

2. If a number is divisible by 4, then it is also divisible by 2.
Solution :
Using Direct Proof :
Assume : x is divisible by 4
Then : x = k * 4 ; where k is some integer (by definition of division)
So, x = k * (2 * 2)
So, x = (k * 2)* 2 (Associative property of multiplication)
So, x = P * 2 where P = k * 2 ; is an integer.
Thus, we can say that x is divisible by 2 also.

3. By using contradiction, prove that : If y + y = y then y = 0.
Solution :
Let P : y +y = y & Q : y = 0
To prove : (P ⋀ ¬Q) is false as (P ⋀ ¬Q) is false ,then¬(P ⋀ ¬Q) is true, and the equivalent statement P ⇒ Q is likewise true.
P : y + y= y and ¬Q : y~= 0.
(P ⋀ ¬Q) means : Then 2y =y and as y ~= 0 we can divide both sides by y.
As a result, we get : 2 = 1, which is a contradiction.
So, (P ⋀ ¬Q) is false and hence P ⇒ Q is true.

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